FINAL EXAMINATION — VERSION B

DevOps, IaC, Kubernetes & Web Application Security
Date: May 14, 2026 | Duration: 3 Hours | Total Marks: 100

Instructions: Answer ALL questions. Part A contains 20 MCQs (40 marks). Part B contains 12 short-answer questions (60 marks). Be precise — full marks require specific terminology, not vague descriptions. Code identifiers, command names, and configuration keys must be spelled correctly.

PART A — MULTIPLE CHOICE QUESTIONS (40 marks · 2 marks each)

1. In a Dockerfile, which instruction sets the working directory inside the container for subsequent RUN, CMD, COPY, and ENTRYPOINT instructions?

A) CHDIR

B) WORKDIR

C) SETDIR

D) PATH

2. A Pod manifest declares `kind: Pod` and `apiVersion: v1`. Inside `spec.containers`, you set `image: nginx:latest`. What does `:latest` guarantee?

A) The most recent stable production release

B) The image tagged 'latest' in the registry — not necessarily the newest version

C) The image will auto-update whenever a new version exists

D) The image is digitally signed by Docker

3. In Terraform, a variable declared with `sensitive = true` will:

A) Encrypt the variable value in the state file

B) Prevent the value from being logged during plan and apply

C) Block the variable from being set via the CLI

D) Require multi-factor authentication to apply changes

4. A Kubernetes Service of type `NodePort` exposes the service on each Worker Node within which port range by default?

A) 1024–8080

B) 8000–9000

C) 30000–32767

D) 80–443

5. Which OWASP vulnerability is BEST illustrated by injecting `& ping -c 10 attacker.com &` into a parameter that gets passed to a shell command?

A) SQL Injection

B) Cross-Site Scripting (XSS)

C) OS Command Injection

D) Server-Side Request Forgery (SSRF)

6. In the OWASP Password Storage Cheat Sheet, what is the PRIMARY purpose of a salt?

A) Make hashes harder to compute (slower)

B) Encrypt the password reversibly

C) Ensure identical passwords produce different hashes, defeating precomputed (rainbow) tables

D) Reduce the size of stored hashes

7. Terraform's `aws_launch_configuration` resource has a known limitation that motivates a specific lifecycle pattern. Which one?

A) It cannot reference variables

B) Launch configurations are immutable; replacing one referenced by an ASG requires create_before_destroy

C) It does not support tags, so it must use ignore_changes

D) It cannot be used with t2.micro instance types

8. Which Kubernetes component on the control plane assigns newly created Pods to Worker Nodes?

A) kubelet

B) kube-proxy

C) Scheduler

D) Controller Manager

9. In Terraform, the syntax `data.aws_vpc.default.id` refers to:

A) The ID attribute of a resource named 'default' of type aws_vpc

B) The ID attribute of a data source named 'default' of type aws_vpc

C) A module output called default.id

D) A local variable defined in defaults.tf

10. An attacker submits `filename=../../../etc/passwd` in a URL parameter and successfully reads the file. This is an example of:

A) SSRF

B) Path Traversal (Directory Traversal)

C) File Upload Vulnerability

D) Open Redirect

11. Which Kubernetes object is responsible for ensuring a stable set of pod replicas is running at any given time, by monitoring desired vs. current state?

A) DaemonSet

B) StatefulSet

C) ReplicaSet

D) Job

12. A Terraform backend configured for S3 with a DynamoDB table serves two purposes. Which pair is correct?

A) Compute caching and credential storage

B) Remote state storage (S3) and state locking (DynamoDB)

C) Log aggregation (S3) and metrics (DynamoDB)

D) Module registry (S3) and provider cache (DynamoDB)

13. In a CI/CD pipeline, the distinction between Continuous *Delivery* and Continuous *Deployment* is:

A) Delivery uses Docker; Deployment uses Kubernetes

B) Delivery requires manual approval to release to production; Deployment releases automatically

C) Delivery is for backend; Deployment is for frontend

D) They are synonymous and interchangeable

14. A Kubernetes `Secret` stores sensitive values. By default, the data in a Secret manifest is:

A) Encrypted with AES-256

B) Encrypted with the cluster's TLS private key

C) Base64 encoded (NOT encrypted)

D) Stored as plain text

15. Which Terraform expression correctly references the public IP of an EC2 instance resource named 'web' in an output block?

A) `${resource.aws_instance.web.public_ip}`

B) `aws_instance.web.public_ip`

C) `var.aws_instance.web.public_ip`

D) `data.aws_instance.web.public_ip`

16. In Cross-Site Scripting (XSS), the attacker's malicious script runs in the victim's browser. Why is the script trusted by the browser?

A) The browser trusts all JavaScript by default

B) The script came from a website the user has visited before

C) The script appears to originate from the trusted target website's domain

D) Browsers cannot detect malicious code

17. Which of these is NOT a Kubernetes health-check probe type?

A) Liveness probe

B) Readiness probe

C) Startup probe

D) Heartbeat probe

18. Why is Docker fundamentally lighter than a Virtual Machine on the same host?

A) Docker images are always smaller than 100 MB

B) Containers share the host OS kernel; VMs each run a full guest OS

C) Docker uses GPU acceleration

D) VMs cannot run Linux

19. In Terraform, the `terraform init` command primarily does which of the following?

A) Applies the infrastructure changes

B) Downloads provider plugins and initializes the backend

C) Destroys the existing infrastructure

D) Validates the syntax of .tf files only

20. An application checks access permissions in an upstream component (e.g., reverse proxy), but the backend service trusts any request from `127.0.0.1`. An attacker exploits this trust by forcing the server to make a request to its own admin endpoint. This attack is:

A) Broken Access Control via SSRF

B) Session Hijacking

C) CSRF (Cross-Site Request Forgery)

D) DNS poisoning

PART B — SHORT ANSWER QUESTIONS (60 marks · 5 marks each)

Keep answers concise. 3–6 sentences each, unless code is requested.

1. Given this Dockerfile snippet, identify TWO mistakes and explain how to fix each:

```
FROM python:3.11-slim
COPY . .
RUN pip install -r requirements.txt
WORKDIR /app
CMD python app.py
```

[5 marks]

2. A developer claims: *"Our CI pipeline runs unit tests on every commit, so we have CI/CD."* Is this statement accurate? Explain what is missing for true CI/CD and define the boundary between CI, Continuous Delivery, and Continuous Deployment.

[5 marks]

3. Describe what happens, step by step, when you run `kubectl apply -f deployment.yaml` **for a new Deployment with** `replicas: 3`**. Mention at least the API server, etcd, scheduler, and kubelet roles.**

[5 marks]

4. Explain the DRY violation in the following Terraform snippet and rewrite it correctly using a variable:

```
resource "aws_security_group" "sg" {
  ingress { from_port = 8080; to_port = 8080; ... }
}
resource "aws_instance" "web" {
  user_data = "busybox httpd -p 8080"
}
```

[5 marks]

5. An attacker injects `' OR '1'='1' --` **into a login form's username field. Show the resulting SQL query (given the original was** `SELECT * FROM users WHERE username='[input]' AND password='[input]'`**) and explain precisely WHY the attacker is authenticated without knowing the password.**

[5 marks]

6. Compare a ConfigMap and a Secret in Kubernetes. Give one valid use case for each, and explain why you should NOT store database passwords in a ConfigMap.

[5 marks]

7. In Terraform, what is the difference between an *implicit dependency* and an *explicit dependency*? Provide a one-line code example of each.

[5 marks]

8. Define SSRF and explain how it bypasses access controls when the application implicitly trusts requests from `localhost` **or** `127.0.0.1`**. Give one mitigation.**

[5 marks]

9. The OWASP Password Storage Cheat Sheet recommends combining salting with a work factor. Explain what each does, and why using ONLY salting (without a slow

hashing algorithm) is insufficient against modern GPU-based attacks.

[5 marks]

10. You discover that `terraform.tfstate` **contains a plaintext database password in a team Git repository. List THREE concrete remediation steps you would take, in order of priority.**

[5 marks]

11. Explain the role of an Ingress Controller in Kubernetes. Why might you choose an NGINX-based Ingress over creating one LoadBalancer Service per microservice in a system with 20 microservices?

[5 marks]

12. A web application encrypts credit card numbers in the database but decrypts them automatically on every query. A SQL injection vulnerability is then discovered. Explain why the encryption did NOT protect the data, and describe TWO design changes that would have prevented exposure.

[5 marks]

— END OF EXAMINATION —

Review your answers before submitting. Partial credit is awarded for partial reasoning. Good luck.